

TASK 3: Low-Level Network Packet Inspection & Protocol Analysis

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vagrant@metasploitable3-ub1404:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:68:3b:19 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.103/24 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe68:3b19/64 scope link
        valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
    link/ether 02:42:3b:a6:94:25 brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
        valid_lft forever preferred_lft forever
    inet6 fe80::42:3bff:fea6:9425/64 scope link
        valid_lft forever preferred_lft forever
vagrant@metasploitable3-ub1404:~$
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(kali㉿kali)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:5a:87:bc brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.10/24 scope global eth0
        valid_lft forever preferred_lft forever

(kali㉿kali)-[~]
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(kali㉿kali)-[~]  
$ sudo ip neigh flush all  
ping -c 4 192.168.56.103  
PING 192.168.56.103 (192.168.56.103) 56(84) bytes of data.  
64 bytes from 192.168.56.103: icmp_seq=1 ttl=64 time=0.547 ms  
64 bytes from 192.168.56.103: icmp_seq=2 ttl=64 time=0.386 ms  
64 bytes from 192.168.56.103: icmp_seq=3 ttl=64 time=0.324 ms  
64 bytes from 192.168.56.103: icmp_seq=4 ttl=64 time=0.305 ms  
  
— 192.168.56.103 ping statistics —  
4 packets transmitted, 4 received, 0% packet loss, time 3058ms  
rtt min/avg/max/mdev = 0.305/0.390/0.547/0.095 ms
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(kali㉿kali)-[~] eth0 host 192.168.56.103 -w /tmp/task3.pcap  
$ nmap -sS -p 21,22,80,445 192.168.56.103  
tcpdump: listening on eth0, link-type EN10MB (Ethernet), snapshot length 2621  
curl -d "username=admin&password=password" http://192.168.56.103/login.php  
  
ftp 192.168.56.103  
# Name: msfadmin  
# Password: msfadmin  
# ls  
# bye  
  
ssh msfadmin@192.168.56.103  
# Password: msfadmin  
# exit  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-29 05:49 -0400  
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers  
Nmap scan report for 192.168.56.103  
Host is up (0.00041s latency).  
  
PORT      STATE SERVICE  
21/tcp    open  ftp  
22/tcp    open  ssh  
80/tcp    open  http  
445/tcp   open  microsoft-ds  
MAC Address: 08:00:27:68:3B:19 (Oracle VirtualBox virtual NIC)
```

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Host is up (0.00041s latency).
Nmap scan report for 192.168.56.103:
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
80/tcp    open  http
445/tcp    open  microsoft-ds
MAC Address: 08:00:27:68:3B:19 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.21 seconds
<!DOCTYPE HTML PUBLIC "-//IETF//DTD HTML 2.0//EN">
<html><head>
<title>404 Not Found</title>
</head><body>
<h1>Not Found</h1>
<p>The requested URL /login.php was not found on this server.</p>
<hr>
<address>Apache/2.4.7 (Ubuntu) Server at 192.168.56.103 Port 80</address>
</body></html>
Connected to 192.168.56.103.
220 ProFTPD 1.3.5 Server (ProFTPD Default Installation) [192.168.56.103]
Name (192.168.56.103:kali): vagrant
331 Password required for vagrant
Password:
230 User vagrant logged in
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> █

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(kali㉿kali)-[~]
└─$ sudo tcpdump -r /tmp/task3.pcap -A 'tcp port 21' | grep -E "USER|PASS"
[sudo] password for kali:
reading from file /tmp/task3.pcap, link-type EN10MB (Ethernet), snapshot length 262144
tcpdump: pcap_loop: truncated dump file; tried to read 66 captured bytes, only got 64
05:48:09.983841 IP 192.168.56.10.53982 > 192.168.56.103.ftp: Flags [P.], seq 1:15, ack 75, win 251, options [nop,nop,TS val 2617677118 ecr 596571], length 14: FTP: USER vagrant
...>... .[USER vagrant
05:48:13.425310 IP 192.168.56.10.53982 > 192.168.56.103.ftp: Flags [P.], seq 15:29, ack 110, win 251, options [nop,nop,TS val 2617680559 ecr 600253], length 14: FTP: PASS vagrant
..... (.PASS vagrant
05:49:24.831930 IP 192.168.56.10.34824 > 192.168.56.103.ftp: Flags [P.], seq 1:15, ack 75, win 251, options [nop,nop,TS val 666074621 ecr 617516], length 14: FTP: USER vagrant
'.}.. l,USER vagrant
05:49:28.377708 IP 192.168.56.10.34824 > 192.168.56.103.ftp: Flags [P.], seq

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14: FTP: USER vagrant
'.}.. l,USER vagrant 0 host 192.168.56.103 -w /tmp/task3.pcap
05:49:28.377708 IP 192.168.56.10.34824 > 192.168.56.103.ftp: Flags [P.], seq
15:29, ack 110, win 251, options [nop,nop,TS val 666078167 ecr 618956], lengt
h 14: FTP: PASS vagrant
'.... q.PASS vagrant
05:51:43.295873 IP 192.168.56.10.50196 > 192.168.56.103.ftp: Flags [P.], seq
1:15, ack 75, win 251, options [nop,nop,TS val 3808759960 ecr 652143], length
14: FTP: USER vagrant
..... .oUSER vagrant
05:51:46.882781 IP 192.168.56.10.50196 > 192.168.56.103.ftp: Flags [P.], seq
15:29, ack 110, win 251, options [nop,nop,TS val 3808763547 ecr 653555], leng
th 14: FTP: PASS vagrant
..... ..PASS vagrant

(kali@kali)-[~]
└─$ sudo tcpdump -r /tmp/task3.pcap -A 'tcp port 80' | grep -E "POST|username
|password|pma_|user=|pass="
reading from file /tmp/task3.pcap, link-type EN10MB (Ethernet), snapshot leng
th 262144
05:47:55.239584 IP 192.168.56.10.45710 > 192.168.56.103.http: Flags [P.], seq
1:190, ack 1, win 251, options [nop,nop,TS val 2532895901 ecr 596569], lengt
h 189: HTTP: POST /login.php HTTP/1.1
..... .YPOST /login.php HTTP/1.1
username=admin&password=password
05:49:19.061497 IP 192.168.56.10.43032 > 192.168.56.103.http: Flags [P.], seq
1:190, ack 1, win 251, options [nop,nop,TS val 1144983098 ecr 617514], lengt
h 189: HTTP: POST /login.php HTTP/1.1

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MT; path=/phpmyadmin/; httponly
05:53:25.538345 IP 192.168.56.10.46366 > 192.168.56.103.http: Flags [P.], seq
  1:180, ack 1, win 251, options [nop,nop,TS val 2473144451 ecr 679102], length
  179: HTTP: POST /drupal/ HTTP/1.1
reading from file /tmp/task3.pcap, link-type EN10MB (Ethernet), snapshot length 262144
POST /drupal/ HTTP/1.1
name=admin&pass=password
<div class="form-item form-type-password form-item-pass">
  <input type="password" id="edit-pass" name="pass" size="15" maxlength="60" class="form-text required" />
<li class="last"><a href="/drupal/?q=user/password" title="Request new password via e-mail.">Request new password</a></li>
<p><strong>What is your password?</strong><br />
tcpdump: pcap_loop: truncated dump file; tried to read 66 captured bytes, only got 64

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(kali㉿kali)-[~]
└─$ sudo tcpdump -r /tmp/task3.pcap 'tcp[tcpflags] & tcp-syn != 0' | head -20

reading from file /tmp/task3.pcap, link-type EN10MB (Ethernet), snapshot length 262144
tcpdump: pcap_loop: truncated dump file; tried to read 247 captured bytes, only got 49
05:46:14.622686 IP 192.168.56.10.33892 > 192.168.56.103.http: Flags [S], seq 266759500, win 64240, options [mss 1460,sackOK,TS val 1207552638 ecr 0,nop,wscale 8], length 0
05:46:15.637526 IP 192.168.56.10.33892 > 192.168.56.103.http: Flags [S], seq 266759500, win 64240, options [mss 1460,sackOK,TS val 1207553652 ecr 0,nop,wscale 8], length 0

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05:47:55.245832 IP 192.168.56.10.53982 > 192.168.56.103.ftp: Flags [S], seq 1
373143133, win 64240, options [mss 1460,sackOK,TS val 2617662380 ecr 0,nop,ws
cale 8], length 0
05:47:55.246196 IP 192.168.56.103.ftp > 192.168.56.10.53982: Flags [S.], seq
3897414052, ack 1373143134, win 28960, options [mss 1460,sackOK,TS val 596570
ecr 2617662380,nop,wscale 7], length 0
05:48:20.692010 IP 192.168.56.10.35276 > 192.168.56.103.ssh: Flags [S], seq 1
814911727, win 64240, options [mss 1460,sackOK,TS val 1957241277 ecr 0,nop,ws
cale 8], length 0

(kali㉿kali)-[~]
└─$ sudo tcpdump -r /tmp/task3.pcap -A 'tcp port 22' | head -30
reading from file /tmp/task3.pcap, link-type EN10MB (Ethernet), snapshot leng
th 262144
05:47:55.180240 IP 192.168.56.10.53815 > 192.168.56.103.ssh: Flags [S], seq 3
472282983, win 1024, options [mss 1460], length 0
E..,Y{..'.H...8
..8g.7.....g....`...+.....
05:47:55.180755 IP 192.168.56.103.ssh > 192.168.56.10.53815: Flags [S.], seq
3379447917, ack 3472282984, win 29200, options [mss 1460], length 0
E..,..@.@.I
..8g..8
...7.nHm...h`.r.....
05:47:55.180768 IP 192.168.56.10.53815 > 192.168.56.103.ssh: Flags [R], seq 3
472282984, win 0, length 0
E..(..@.@.I...8
..8g.7.....h....P...Gq..
05:48:20.692010 IP 192.168.56.10.35276 > 192.168.56.103.ssh: Flags [S], seq 1

```